Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/18/2005
PI Name:	Batt, Carl A. Ph.D.		
Project Title:	Biodegradable Polymers		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	09/01/2003	**End Date:**	03/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2943		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth"s gravity		

A novel class of biodegradable nanocomposites will be developed and the underlying physical basis for their unique structure-function relationship will be elucidated. As a model system we will explore the polyhydroxyalkanoates for which significant knowledge on the genetics and biochemistry exists. The nanocomposites will be made by incorporating inorganic nanoparticles into the polymer matrix. The nanocomposites will be characterized and their properties will be evaluated. The physical basis through which the functional properties are altered is not clear and part of this effort will seek to gain new knowledge on the interactions at the polymer-inorganic interface. We will apply state-of-art molecular biology to create biopolymers with unique and tailored composition. Further we will partner in an interdisciplinary effort that will join material scientists, nanofabrication and biologists into the design, engineering, production and testing of the system. Biodegradable nanocomposite polymers will be developed and their ability to serve as a base material for microfabrication documented. 1)understand how the physical (i.e. permeability) and

Task Description:	mechanical properties of the nanocomposites are affected by nanostructure (e.g. intercalated vs. delaminated hybrids) and strength of the polymer-silicate interface (by varying the interaction between the polymer and the silicate surface) 2)use this understanding to rationally design and synthesize nanocomposites with enhanced properties. 3)Develop novel biopolymer components 4)Integrate silicates into biopolymers creating novel nanocomposites and test their utility for fabrication of nanosize features Biopolymers are 'green' and can have a significant role as materials for long term space flight reducing the environmental impact and potentially being a major structural component in a closed system. Their physical properties are however sometimes not optimal for a wide variety of applications. Reinforcement of polymers with a second phase inorganic or organic, with fibers or particles to produce polymer composites is common in the production of modern plastics. Polymer nanocomposites represent a radical alternative to conventional polymer composites. Because of their nanometer phase dimensions the nanocomposites exhibit properties dramatically different from their micron-scale counterparts. These include enhanced mechanical properties, decreased gas permeability, increased solvent resistance and decreased flammability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our method of chemical modifications of biopolymer can benefit the development of similar functionalized polymers that can serve as environmentally friendly matrices for novel nanocomposite materials.
Task Progress:	At current stage, we have successfully developed a novel hybrid polymeric nanocomposite based on PHAs biopolymer. In this project, PHA-PDMS hybrids were created through a crosslinking reaction between functional groups present in the PHAs and complementary groups present in the PDMS. Fumed silica was also used as a compatibilizer and a filler in the system. PDMS and PHA are incompatible polymer blends. Therefore, the presence of fumed silica helps the hybrid PDMS/PHA elastomer to become homogenous. As a filler, fumed silica acts as physical crosslinks for the elastomer network. This hybrid material represents the first of its kind reported to date, which has properties quite similar to standard PDMS rubber, but unlike such PDMS, this material is potentially easier to be degraded compared to conventional PDMS. We are also investigating the feasibility of using this novel PHA-PDMS hybrid nanocomposite for constructing microfluidic networks on chip-based biosensor devices.
Bibliography Type:	Description: (Last Updated: 09/03/2009)
Articles in Peer-reviewed Journals	Niamsiri N, Delamarre SC, Kim YR, Batt CA. "Engineering of chimeric class II polyhydroxyalkanoate synthases." Appl Environ Microbiol. 2004 Nov;70(11):6789-99. PubMed PMID: 15528546 , Nov-2004